

Module 1: Introduction to Microprocessors & Embedded Systems

Definition of Microprocessor & Embedded System

Microprocessor:

- A **microprocessor (MPU)** is an integrated circuit (IC) that functions as the **CPU of a computer system**.
- It performs arithmetic operations, logic operations, and manages data flow between memory and I/O devices.
- **Example:** Intel 8086 used in early personal computers.

Components of a Microprocessor:

1. **ALU (Arithmetic Logic Unit):** Performs arithmetic (add, subtract) and logical operations (AND, OR).
2. **Registers:** Small, high-speed storage locations for temporary data.
3. **Control Unit (CU):** Directs operations of the processor.
4. **Buses:** Pathways for data, address, and control signals.

Embedded System:

- A **special-purpose computer system** designed to perform **dedicated tasks** within a larger system.
- Can be **real-time** (response must happen within a time limit) or **non-real-time**.
- **Example 1:** A washing machine controller (controls water level, drum speed, temperature).
- **Example 2:** An automotive airbag system (activates within milliseconds during a crash).

Characteristics:

- Dedicated function, compact, low-power, real-time operation in many cases.

History & Evolution of Microprocessors (Intel 4004 → Intel Core i9)

1. Intel 4004 (1971)

- **Bit-width:** 4-bit
- **Clock speed:** 740 kHz

- **Transistors:** 2,300
- **Memory support:** 640 bytes
- **Significance:** First commercially available microprocessor; primarily used in calculators like Busicom 141-PF.
- **Architecture:** Single-chip CPU capable of executing 46 instructions.

2. Intel 8008 (1972)

- **Bit-width:** 8-bit
- **Clock speed:** 0.5–0.8 MHz
- **Transistors:** 3,500
- **Applications:** Early terminals and industrial controllers
- **Improvement:** More data width (8-bit) allowing better arithmetic and logic operations.

3. Intel 8080 (1974)

- **Bit-width:** 8-bit
- **Clock speed:** 2 MHz
- **Memory:** 64 KB addressable
- **Applications:** Altair 8800 microcomputer, first widely used personal computer.
- **Significance:** First processor widely adopted for hobbyist and personal computing.

4. Intel 8086 (1978)

- **Bit-width:** 16-bit
- **Clock speed:** 5–10 MHz
- **Architecture:** Segmented memory model, instruction pipelining
- **Significance:** Foundation for the x86 architecture, still used in modern PCs.
- **Applications:** IBM PC (1981), early personal computers.

5. Intel 80286 (1982)

- **Bit-width:** 16-bit
- **Clock speed:** 6–25 MHz
- **Memory:** Supports up to 16 MB
- **Features:** Protected mode, multitasking, backward compatible with 8086
- **Applications:** IBM PC/AT, business computers.

6. Intel 80386 (1985)

- **Bit-width:** 32-bit
- **Clock speed:** 12–40 MHz
- **Features:** Virtual memory support, multitasking, 32-bit registers
- **Applications:** PCs running Windows 3.x, UNIX systems.

7. Intel 80486 (1989)

- **Bit-width:** 32-bit
- **Clock speed:** 20–100 MHz
- **Features:** Integrated FPU, pipelined architecture, 8 KB on-chip cache
- **Applications:** High-performance desktops, early workstations

8. Intel Pentium Series (1993–2000)

- **Pentium (P5, 1993):** Superscalar, pipelined, enhanced FPU, 60–300 MHz
- **Pentium Pro (1995):** Out-of-order execution, 32-bit, enterprise/server use
- **Pentium II (1997):** Multimedia (MMX instructions), 233–450 MHz
- **Pentium III (1999):** SSE instructions, 450–1400 MHz
- **Pentium 4 (2000):** NetBurst microarchitecture, high clock speed (1.3–3.8 GHz).

9. Intel Core Series (2006–Present)

- **Core 2 Duo / Quad (2006):** Multi-core, energy-efficient, 64-bit
- **Core i3 / i5 / i7 / i9 (2008–Present):**
 - Multi-core, Hyper-Threading, Turbo Boost
 - L3 cache integrated
 - Support for virtualization
 - i9: High-end, 6–24 cores, 3–5 GHz, gaming & workstation CPUs
- **Applications:** Modern laptops, desktops, servers, gaming rigs, AI processing

Key Evolution Highlights

1. **Bit-width expansion:** 4-bit → 8-bit → 16-bit → 32-bit → 64-bit
2. **Clock speed growth:** 0.74 MHz (4004) → 5 GHz+ (modern Core i9)
3. **Integration & architecture:** From single-chip CPU → multi-core, integrated caches, FPU, GPU support

4. **Applications:** Calculators → PCs → Laptops → Gaming & AI Workstations
5. **Instruction set evolution:** Basic arithmetic → SIMD (MMX/SSE/AVX) → Advanced multi-threading & virtualization

Example in Real Life:

- Early calculators and simple PCs used **4004 and 8080**.
- Modern PCs and laptops use **Pentium or Core i-series processors**.
- Embedded systems in cars might use microcontrollers derived from **8086/ARM architecture**.

Difference: Microprocessor vs. Microcontroller

Feature	Microprocessor (MPU)	Microcontroller (MCU)
Definition	CPU on a chip, requires external peripherals	CPU + RAM + ROM + I/O on a single chip
Focus	General-purpose computing	Specific embedded tasks
Peripherals	Needs external memory and I/O devices	On-chip memory and I/O
Cost	Higher for small applications	Lower, compact for embedded systems
Power Consumption	Higher	Lower, often battery-operated
Example	Intel 8086, Pentium	Atmega328, PIC16F877, STM32

Applications of Embedded Systems

Embedded systems are everywhere in modern life.

1. Automotive Applications

Embedded systems are widely used in modern vehicles to improve safety, comfort, and performance.

- **Examples:**
 - **Anti-lock Braking System (ABS):** Monitors wheel speed and controls braking to prevent skidding.
 - **Airbag System:** Detects collisions and deploys airbags instantly.

- **Engine Control Unit (ECU):** Optimizes fuel injection and engine performance.
- **Infotainment Systems:** Touchscreen entertainment and navigation systems.

2. Consumer Electronics

Embedded systems make devices smarter, smaller, and energy-efficient.

- **Examples:**
 - **Smart TVs:** Embedded software handles display, streaming, and connectivity.
 - **Washing Machines:** Microcontrollers manage water level, temperature, and cycles.
 - **Microwave Ovens:** Timer and power control are managed by embedded controllers.
 - **Digital Cameras:** Auto-focus, image processing, and storage are controlled by embedded systems.

3. Medical Applications

Embedded systems improve diagnosis, treatment, and monitoring.

- **Examples:**
 - **Pacemakers:** Monitors heart rate and delivers electrical impulses to regulate heartbeat.
 - **Medical Imaging Devices:** MRI, CT, and X-ray machines rely on embedded systems for processing.
 - **Patient Monitoring Systems:** Track vital signs like heart rate, blood pressure, and oxygen levels.
 - **Insulin Pumps:** Deliver precise insulin doses based on glucose measurements.

4. Industrial Automation

Embedded systems help in controlling machinery and improving efficiency in industries.

- **Examples:**
 - **Robotic Arms:** Used in assembly lines for precision tasks.
 - **CNC Machines:** Embedded controllers manage cutting, shaping, and drilling.
 - **PLC (Programmable Logic Controllers):** Control factory processes and production lines.
 - **Smart Sensors:** Monitor temperature, pressure, and flow in industrial equipment.

5. Telecommunication

Embedded systems support communication networks and devices.

- **Examples:**
 - **Routers and Modems:** Handle data routing and signal processing.
 - **Mobile Phones:** Embedded software manages calls, apps, and connectivity.
 - **Base Stations:** Control communication between mobile devices and network towers.

6. Aerospace and Defense

Critical systems in defense and aerospace rely heavily on embedded systems.

- **Examples:**
 - **Avionics Systems:** Flight control, navigation, and communication in aircraft.
 - **Missile Guidance Systems:** Real-time calculations for targeting and trajectory.
 - **Drones (UAVs):** Embedded systems control flight, stabilization, and cameras.

7. Home Automation and IoT

Embedded systems enable “smart” homes and connected devices.

- **Examples:**
 - **Smart Thermostats:** Adjust temperature based on occupancy and preferences.
 - **Smart Lights:** Controlled via apps or voice commands.
 - **Security Systems:** Embedded cameras, sensors, and alarms for surveillance.
 - **Voice Assistants:** Devices like Amazon Echo and Google Home.